

STRATEGIC ENVIRONMENTAL RESEARCH AND DEVELOPMENT PROGRAM (SERDP) BROAD AGENCY ANNOUNCEMENT FOR ENVIRONMENTAL RESEARCH AND DEVELOPMENT

Reference: Broad Agency Announcement (BAA), 22 August 2025
U.S. Army Corps of Engineers, Humphreys Engineer Center Support Activity

1. INTRODUCTION

The Strategic Environmental Research and Development Program (SERDP) is the Department of Defense's (DoD's) environmental research and development program, planned and executed in partnership with the Department of Energy and the Environmental Protection Agency. SERDP's role is to fund research and development that addresses innovative technologies or methods that enhance military readiness, improve installation resilience, and strengthen America's defense infrastructure. The development and application of innovative technologies supports operational effectiveness, improves warfighter capabilities, and reduces unnecessary regulatory burdens. Within its broad areas of interest, the Program focuses on Environmental Restoration, Munitions Response, Resource Conservation and Resilience, and Weapons Systems and Platforms. SERDP funds research and development programs in basic and applied research as well as advanced technology development.

1.1. GENERAL INFORMATION FOR PROPOSERS

SERDP seeks white papers from private sector organizations proposing innovative, cost-effective solutions to address DoD's most pressing infrastructure resilience and sustainment challenges. Technologies should have the potential to significantly enhance mission assurance, reduce logistical vulnerabilities, and increase operational efficiency. Topic areas defined by this Broad Agency Announcement (BAA) are located in Appendix A of this document. This BAA is for Private Sector Organizations.

Awardees under this BAA will be selected through a multi-stage review process, including a brief white paper and a full proposal. Some proposals may require an oral presentation to the SERDP Technical Review Board (TRB) for final approval. To be eligible for consideration, proposers must submit a white paper. Any white paper submitted shall be in response to only one of the topic areas set forth in this announcement. White papers marked with ITAR, EAR, a limited distribution statement, Controlled Unclassified Information (CUI), or as company proprietary are not accepted. Please note, only government employees or Department of Defense contractors who have signed non-disclosure agreements have access to SERDP proposals. Proposers may respond to more than one topic area with separate white papers.

After evaluation of white papers, SERDP will contact all proposers and either request or not request each to submit a full proposal. At that time, SERDP will provide detailed instructions for the full proposal format. Full proposals may not be submitted outside the white paper process. Any full proposal that has not been reviewed in the white paper phase will not be evaluated nor considered for award under this BAA. Due to the volume of white papers anticipated, SERDP will not provide debriefs on those that are not requested to submit a full proposal.

Based on an evaluation of the written full proposal, and pending approval by the TRB, SERDP will notify each proposer as to whether the Government Contracting Officer wishes to enter into negotiation for the award of a contract. Proposers are advised that only the Contracting Officer is

legally authorized to bind the Government. SERDP reserves the right to recommend for award any, all, or none of the proposals received. SERDP also reserves the right to recommend a portion of the work proposed in any single proposal for award. There is no commitment by SERDP to make any recommendations for contract awards, nor to be responsible for any money expended by the proposer before contract award is made. It is the sole responsibility of the proposer to make certain the proposal is properly received by SERDP.

The SERDP Office manages the BAA solicitation along with the U.S. Army Corps of Engineers, Humphreys Engineer Center Support Activity (HECSA) Contracting Center in Alexandria, Virginia. For contracting or small business information regarding this BAA solicitation, contact Shannon Benson: 703-428-7407, Shannon.N.Benson@usace.army.mil or the HECSA Small Business Representative, Monique Holmes: 202-934-1816, HECSASmallBusinessProgram@usace.army.mil. General SERDP procedural questions may be referred to the SERDP office at 571-372-6565. For technical information regarding a topic area, contact the individual listed in Appendix A.

1.2. EVALUATION SCHEDULE

This BAA is open for approximately one year from the date of release or until it is replaced by a subsequent announcement.

White papers will not be evaluated against each other since they are not submitted in accordance with a common work statement. SERDP's intent is to review white papers as soon as possible after they are submitted; however, white papers may be reviewed periodically for administrative reasons. Proposers will be notified on the decision regarding their white paper as soon as possible following review.

Full proposals will be evaluated as soon as possible after they are submitted. Review of full proposals may require input from external peer reviewers or other technical experts. Proposals may be required to brief the SERDP TRB depending on their size and scope. TRB meetings are typically held four times each fiscal year in October, March, June, and September.

Contracts are anticipated to be awarded three to six months following final review and approval of each proposal, depending on availability of funding.

2. WHITE PAPER INSTRUCTIONS

2.1 WHITE PAPER LENGTH AND FORMAT

White papers shall be no longer than two (2) pages, type face not less than 11-point, and margins not less than one inch on all sides. All white papers shall be submitted as a single PDF file containing all sections outlined below.

2.2 WHITE PAPER CONTENT

White papers must describe the merits and objectives of the proposer's project in response to the respective topic area. The white paper should concisely describe the following:

- 1. Proposal Title**
- 2. Lead Principal Investigator**
- 3. Lead Organization**
- 4. Objectives:** State the proposed objectives and how the project is responsive to the objectives articulated in the topic area.
- 5. Approach:** Describe the technical approach and methods, preferably structured in hypothesis-driven tasks that clearly identify how the objectives of the proposed project will be addressed. This section should be the primary focus of the white paper.
- 6. Cost Estimate:** The estimated total costs by year. A detailed breakout of costs is not required or desired in the white paper. A detailed cost breakout will be provided in the full proposal, if requested. If selected for funding, proposers will be required to provide a certificate of current cost or pricing data prior to award in accordance with Federal Acquisition Regulation (FAR) 15.403-4(a)(1) if the total contract value is expected to exceed \$2,000,000.00 to the cognizant contracting office.
- 7. Literature Citations:** Provide literature citations for any material cited.

3. SUBMITTAL

White papers are submitted via the SERDP and ESTCP Management System (SEMS). No electronic mail, faxed, or hard copy white papers will be accepted.

Complete all steps below in order to submit a white paper.

- Create a single PDF file containing all required sections outlined in the white paper guidance.
- Click the link below that corresponds to the Program Area the white paper responds to (create a SEMS account if needed):
 - [Environmental Restoration](#)
 - [Munitions Response](#)
 - [Resource Conservation and Resilience](#)
 - [Weapons Systems and Platforms](#)
- Enter all required white paper details into SEMS. If the white paper is associated with an existing SERDP project, enter the project number on the first page of the white paper submission wizard in SEMS. White paper details may be saved and edited prior to final submission.
- Before submitting, add the system-generated proposal number (displayed in front of the white paper title on the “Manage Proposals” page), project title, lead PI name and organization to first page of white paper as instructed in Section 2.2 above.
- Upload the final white paper file.
- Submit the white paper. Only white papers that have been submitted will be considered. White papers with a “Saved” status will not be reviewed.

Edits cannot be made once the white paper has been submitted. SEMS will display an on-line confirmation message and send an email notification to the proposer. It is the sole responsibility of the proposer to make certain the white paper is properly received by SERDP. For questions, contact the SERDP Office at 571-372-6565.

Note: A signed cover page is **not required** for white papers.

4. EVALUATION FACTORS FOR WHITE PAPERS AND FULL PROPOSALS

The following evaluation factors will be the sole basis for reviewing white papers and full proposals submitted in response to this BAA. SERDP relevance is a pass/fail criteria evaluated at the white paper stage only; white papers not passing this gate will not be further evaluated. Among the other evaluation factors for both white papers and full proposals, Technical Merit is more important than Personnel, Cost, and Transition Plan.

SERDP RELEVANCE (PASS/FAIL)

Proposal reviewers will assess whether the white paper submission (1) responds to the objectives as described in the topic area, and (2) falls within the SERDP mission to support basic and applied research or advanced technology development. SERDP Relevance is a threshold review and if the white paper is determined to be not relevant, no further evaluation of the white paper will be conducted.

TECHNICAL MERIT

The overall scientific and technical merit of the white paper must be clearly identifiable. The evaluation will consider the proposed approach and its substantiation by calculations, test data, and references. Emphasis will be placed on the proposer's demonstration of a thorough understanding of the environmental issue. The proposer must demonstrate the ability to execute the work by providing a comprehensive, logical, orderly, and concise plan that indicates major tasks, milestones, critical paths, go/no-go decision points and key events leading to the completion of the project in the proposed time frame. In addition, the proposer must show how the technical approach and proposed tasks will address the project objectives. Strong consideration will be given to innovation; however, the degree of risk associated with individual proposals will be weighed against potential benefits. The proposal should clearly articulate how the research will advance the state of the science.

PERSONNEL (FULL PROPOSALS ONLY)

Proposal reviewers will examine and assess the applicable qualifications, capabilities, demonstrated achievements, and proposed time commitment to the project by the proposed principal(s) and other key personnel.

COST (FULL PROPOSALS ONLY)

Proposal reviewers will consider the reasonableness of the proposed cost, as well as the appropriateness and substantiation of costs for the technical complexity described. Cost sharing or leveraged resources also will be considered.

TRANSITION PLAN (FULL PROPOSALS ONLY)

The transition plan of the proposed research product(s) should demonstrate a clear understanding of how the project's results will transition to implementation either directly through future demonstrations or through future development and show a linkage between the work proposed and the needs of ultimate end user of the results. Coordination between the proposer and targeted end user community is of value for late-stage development projects.

APPENDIX A: SERDP TOPIC AREAS

ENVIRONMENTAL RESTORATION

SERDP's Environmental Restoration Program Area focuses on restoring and managing impacted lands on current and former military installations. The program supports the development of innovative technologies to assess, remediate, and manage chemicals of concern in soil, sediments, and water sources. The priority is on practical, field-ready approaches that enhance operational flexibility and reduce long-term costs and obligations for the DoD.

To provide strategic guidance for future research, SERDP and ESTCP periodically hold workshops to identify and prioritize research needs on specific topics. Proposers are strongly encouraged to refer to the Summary Report listed below.

- [Summary Report](#) from the November 2022 SERDP and ESTCP *Strategic Workshop on Management of PFAS in the Environment*

Data needs identified during workshops are relevant to several of the specific areas of interest listed below. Current and anticipated areas of focus include:

Characterization and Monitoring

Development of methodologies or technologies that improve the efficiency and accuracy of long-term monitoring for chemicals of concern or biogeochemical indicators in soils, sediments, and water. Solutions should emphasize rapid deployment, cost savings, and operational readiness.

Reduction in Cost to Complete

Improved understanding of the processes that reduce costs and timelines associated with remediation while ensuring compliance with DoD mission requirements for impacted groundwater or aquatic sediments. Also of interest is the development of innovative tools and technologies that streamline performance assessments and optimize treatment for these matrices. Sites impacted by PFAS, chlorinated solvents, munitions constituents, PCBs, and PAHs are of primary concern.

Reduce Source Loading of Munitions Constituents

Improved understanding of processes associated with the management of munitions constituents on DoD testing and training ranges throughout the lifecycle of their use. Development of technologies that reduce source loading of munitions constituents during routine DoD operations and demilitarization activities is also of interest. The goal is to enhance sustainability and operational efficiency while minimizing environmental impact. Proposers are encouraged to refer to the [Summary Report](#) from the July 2015 SERDP and ESTCP *Workshop on Research and Demonstration Needs for Management of Munitions Constituents*.

Stormwater Treatment

Development of cost-effective and low-maintenance solutions are sought for managing stormwater runoff from DoD facilities. Technologies should integrate with existing Best Management Practices and support compliance with National Pollution Discharge Elimination System (NPDES) permit requirements while minimizing regulatory burden.

Wastewater Treatment

Development of decentralized wastewater treatment and recycling systems that are energy efficient, low maintenance, and capable of supporting fixed installations. Solutions that operate in an energy neutral configuration or produce materials that can be converted into power are of interest.

Risk Assessment

Enhanced understanding of processes that improve the military's ability to assess and predict human and ecological risks from chemicals of concern. Areas of focus include fate and transport of chemicals of concern, exposure pathways to humans, ecological receptor impacts, and tools for assessing environmental risks.

Point of Contact

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MUNITIONS RESPONSE

SERDP's Munitions Response program area supports the development of innovative technologies that can characterize, remediate, and manage sites affected by military munitions with a focus on cost-effective and scalable solutions. Ensuring national security while maintaining environmental and public safety is a priority. The program seeks technologies that enhance operational efficiency, reduce remediation costs, and improve detection and classification capabilities.

Many affected sites contain munitions in underwater environments, with depths typically less than 5 meters, though concerns extend to depths of up to 35 meters. These environments include ponds, lakes, rivers, estuaries, and coastal or open ocean areas. Munitions of interest range from small projectiles and mortars to large bombs.

Current research priorities include:

Wide Area and/or Detailed Survey Techniques

Technologies for conducting large-scale, cost-effective surveys to identify munitions concentrations and clear areas continue to be of interest, but the program has developed a complete suite of physics-based systems and the interest has shifted to maintaining an awareness of scientific advances for improvement. Sensor modalities should maximize coverage rates while maintaining reliable probabilities of detection and classification.

Cost-Effective Recovery and Disposal Methods

Improved recovery and disposal techniques are required to efficiently and safely remove munitions from the underwater environment. Current practices relying on divers for manual retrieval are costly and hazardous; the program seeks innovative recovery technologies suited for shallow water environments, particularly where munitions may pose a risk to the public, such as areas frequented by recreational divers. Proposed methods must address explosive safety concerns and provide cost-effective, safe, and environmentally responsible remediation solutions. Technologies that reduce reliance on blow-in-place operations to mitigate impacts on marine life are of particular interest.

Mobility of Munitions in the Dynamic Underwater Environment

The search for a limited set of parameters that describe water column dynamics, sediment physical properties and munition configurations, and in turn, predict transition from stationary to mobile munition behavior and/or burial potential remains of high interest.

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RESOURCE CONSERVATION AND RESILIENCE

SERDP's Resource Conservation and Resilience (RC) program area supports research that enhances DoD mission readiness through the demonstration and evaluation of innovative, cost-effective technologies for support of installation natural infrastructure and efficient compliance with legal requirements. This program prioritizes technology that sustains training and testing capabilities, improves operational effectiveness, and meets long-term resource management needs. Research efforts should focus on optimizing installation management for long-term military use to enhance national defense priorities by efficiently meeting legal obligations and preventing future regulatory impacts through science and proactive management.

Managing Wildland Fire

Technologies are sought to improve the management of prescribed fires on and around DoD installations to ensure training and testing activities remain uninterrupted while preventing wildfire impacts on surrounding communities. Solutions should focus on tools that improve safe and effective use of prescribed fire for installation needs, reduce the risk of uncontrolled wildfire, and minimize smoke exposure to nearby populations. Research should prioritize practical, field-ready approaches that improve fire management without increasing operational costs or logistical burdens.

Natural Resources

Natural resource science and technology are sought to support the management of land, water, and species that are essential to DoD installations and their operational readiness. Solutions should preserve realistic and unimpeded training and testing lands while supporting resource management that enhances long-term operational effectiveness. Key areas of interest include technologies that optimize land use, sustain range viability, and manage at-risk or invasive species that may impact operations. Research should focus on scalable, low-maintenance solutions that enhance land stability without increasing costs or administrative burdens.

Resilience of Built and Natural Infrastructure

Resilience research focuses on developing and demonstrating advanced tools, technologies, and methods for maintaining natural and built DoD infrastructure to support capabilities, training, and testing of new missions. The program focus is on research that examines natural and built systems challenged by weather extremes, compound natural hazards, identification and accurate prediction of rapid transformation in installation environments, and identification of management responses needed to maintain conditions for military uses. Forensic analysis of recent or historical regime shifts impacting natural and built systems is also of interest for improving near term forecasting methods.

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WEAPONS SYSTEMS AND PLATFORMS

SERDP's Weapons Systems and Platforms (WP) program area supports the development of innovative technologies and materials to ensure America's warfighters have access to current critical materials, next-generation materials, and manufacturing and repair processes to maximize the lethality and readiness of our weapons systems and platforms from production through decommissioning.

Manufacturing and Maintenance

Development of technologies that enable cost-effective maintenance and modernization of military platforms are of interest. Areas of focus include advanced coatings, primers, and surface treatments that improve durability and reduce corrosion, as well as alternative solvents and technologies for efficient material processing. Additional priorities include high-performance materials for additive manufacturing and advanced joining techniques to enhance structural integrity. Solutions should emphasize increased longevity and reduced sustainment costs for DoD assets while minimizing the need for costly protective measures such as extensive PPE and engineering controls. Development of non-flammable refrigerants and novel refrigeration technologies that do not use refrigerants are of significant interest to ground vehicles. The use of alternative flame retardants for soldiers, vehicles, and support equipment applications are of interest. Alternate chemical treatments and materials for soldier uniforms and equipment with commercially viable materials and processes are of interest. Proposed technologies should minimize or eliminate the use of TSCA-restricted chemicals, hazardous chemicals, and emerging chemicals of concern. Proposals should hypothesize fundamental reasons for the efficacy of the proposed alternative materials, processes, or technology and propose methods to assess their effectiveness.

To ensure alignment with DoD priorities, SERDP and ESTCP periodically conduct workshops to identify and prioritize research needs. Proposers are encouraged to refer to the ASETSDefense workshop reports for additional insight.

Energetic Materials and Formulations

Development of alternative materials or fabrication methods for military propellants, explosives, and pyrotechnics are sought. Alternative materials or methods should improve performance, enhance safety, and reduce costs associated with operational requirements and waste disposal. The program is also interested in innovative demilitarization techniques that improve disposal efficiency without compromising mission capability. Solutions should streamline disposal processes, lower costs associated with waste byproducts handling, and ensure compliance with DoD operational requirements with minimal regulatory burden. Proposed technologies should minimize or eliminate the use of TSCA-restricted chemicals, hazardous chemicals, and emerging chemicals of concern. Proposals should hypothesize fundamental reasons for the efficacy of the proposed alternative materials, processes, or technology and propose methods to assess their effectiveness.

Additional information on our current and past efforts in energetic materials is available on our website.

Firefighting

Development of fluorine-free surfactants, additives, and formulations for aqueous film forming foams (AFFF) for firefighting class B fuel fires for shipboard use with saltwater dilution are sought. Technologies should enhance fire suppression performance and other requirements as outlined in MIL-PRF-32725 while using saltwater dilution until a new fluorine-free specification is created for shipboard use. Alternative technologies are also sought to enable fire suppression within the cabin of vehicles, ships, and aircraft. Alternative fire suppressants and dispersion/distribution technologies will be considered with a strong theoretical basis for their selection. Proposed technologies should minimize or eliminate the use of TSCA-restricted chemicals, hazardous chemicals, and emerging chemicals of concern. Proposals should hypothesize fundamental reasons for the efficacy of the proposed alternative process or technology and propose methods to assess its effectiveness.

Critical Minerals, Chemicals, and Materials

Various critical minerals, chemicals, and materials are used on weapons systems and platforms, including, but not limited to neodymium, samarium, graphite, and carbon fiber. Technologies are sought to develop recovery processes for critical minerals and materials from DoD materiel that is not currently recycled and to identify improved recycling technologies and processes to increase recovery efficiency. Extraction and refining technologies with increased effectiveness are sought.

Alternative to critical minerals, materials, and chemicals, including those restricted by TSCA regulations, for application to DoD materiel to improve resilience are sought. Proposed technologies should minimize or eliminate the use of TSCA-restricted chemicals, hazardous chemicals, and emerging chemicals of concern. Proposals should hypothesize fundamental reasons for the efficacy of the proposed alternative materials, processes, or technology and propose methods to assess their effectiveness.

Point of Contact

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